

Thm:

$$H(A(P), t) = \frac{h^*(t)}{(1-t)^{d+1}}$$

$$d = \dim P$$

h^* is a polynomial
of degree $\leq d$

Proof: First assume P is a simplex.

As in the previous proof, every point in $C(P) \cap \mathbb{Z}^{n+1}$
can be written uniquely as

$$\underbrace{\alpha}_{\in D \cap \mathbb{Z}^{n+1}} + \sum_{i=1}^{d+1} q_i (v_i, 1) \quad q_i \in \mathbb{Z}_{\geq 0}$$

Thus

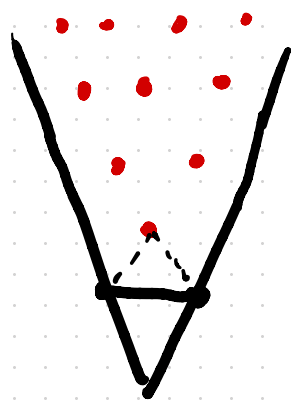
$$H(\underline{k}[x^{\vee_i}y : i=1, \dots, d+1], t)$$

$$H(A(\rho), t) = \sum_{\alpha \in D \cap \mathbb{Z}^{n+1}} t^{\deg \alpha} \cdot \frac{1}{(1-t)^{d+1}}$$

$$= \frac{h_0^* + h_1^* t + \dots + h_d^* t^d}{(1-t)^{d+1}}$$

where h_i^* is the number of integer points in D at height i .

Let $A(P^\circ)$ be the ideal of $A(P)$ generated by monomials $x_1^{a_1} \cdots x_n^{a_n} y^b$ where $(a_1, \dots, a_n, b)$ is in the *relative interior* of $C(P)$.



$C(P)$

Claim: If P is a simplex,

$$H(A(P^\circ), t) = \frac{p(t)}{(1-t)^{d+1}} \quad \begin{array}{l} p \text{ a} \\ \text{polynomial} \end{array}$$

Let $v_1, \dots, v_{d+1}$ be the vertices of P .

$$\text{Let } D^\circ = \left\{ \sum_{i=1}^{d+1} r_i (v_{i,1}) : 0 < r_i \leq 1 \right\}$$

Fact: Every point in $\text{rint } C(P) \cap \mathbb{Z}^{n+1}$ can be written uniquely as

$$\alpha + \sum_{i=1}^{d+1} q_i (v_i, 1) \quad q_i \in \mathbb{Z}_{\geq 0}$$

$$\in \overline{D^0} \cap \mathbb{Z}^{n+1}$$



Therefore,

$$H(A(P^0), t) = \sum_{\alpha \in \overline{D^0} \cap \mathbb{Z}^{n+1}} t^{\deg \alpha} \cdot \frac{1}{(1-t)^{d+1}}$$

$$= \frac{g_0 + g_1 t + g_2 t^2 + \dots + g_{d+1} t^{d+1}}{(1-t)^{d+1}}$$

where g_i is the number of elements of $D \cap \mathbb{Z}^{n+1}$ at height i .

We have $g_i = h_{d+1-i}^*$

We have a bijection $\text{cl } D \rightarrow \text{cl } D$ given by

$$\sum_{i=1}^{d+1} r_i(v_i, 1) \mapsto \sum_{i=1}^{d+1} (1-r_i)(v_i, 1)$$

This sends elements of $\mathcal{O}^\circ \cap \mathcal{D}^{n+1}$ at height i to elements of $\mathcal{O} \cap \mathcal{D}^{n+1}$ at height $d+1-i$.

Therefore, if P is a simplex

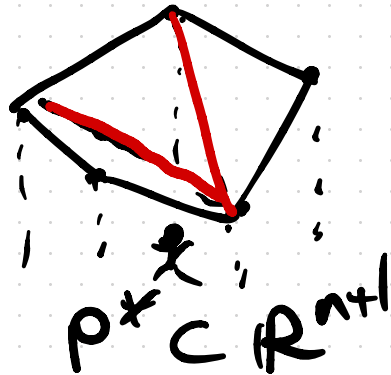
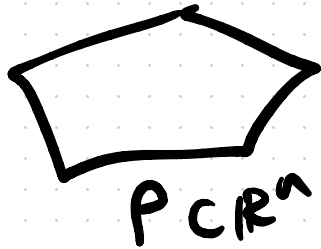
$$H(A(P^0), t) = \frac{h_d^* t + h_{d-1}^* t^2 + \dots + h_0^* t^{d+1}}{(1-t)^{d+1}}$$

or

$$H(A(P^0), t) = (-1)^{d+1} H(A(P), \frac{1}{t})$$

Prop: Every polytope has a triangulation without adding new vertices.

Proof: Choose a random real number for each vertex of the polytope, and lift the vertices to the height of that number



If I stand below the lifted polytope and look up, I see a triangulation of the original polytope.

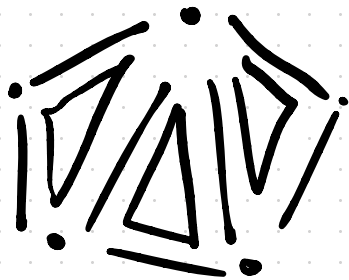
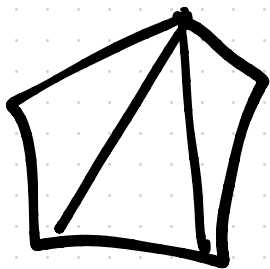
Thm:

$$H(A(P), t) = \frac{h^*(t)}{(1-t)^{d+1}}$$

$$d = \dim P$$

h^* is a polynomial
of degree $\leq d$

Proof: Consider a triangulation T of P whose vertices are all in $\mathbb{Z}^n$. The relative interiors of the simplices of T partition P .



Therefore, the cones over these relative interiors partition $C(P)$.

$$C(P) = \bigsqcup_{\sigma \in T} \text{rint } C(\sigma)$$

Include $\sigma = \phi$,

$$C(\phi) = \{0\}$$

$$\text{rint } C(\phi) = \{0\}$$

$$H(A(P), t) = \sum_{\sigma \in T} H(A(\sigma^o), t)$$

$$= \sum_{\sigma \in T} \frac{g_{\sigma}(t)}{(1-t)^{\dim \sigma + 1}}$$

where g_{σ} is a polynomial of degree $\dim \sigma + 1$ and leading coefficient is 1.

$$= \frac{1}{(1-t)^{d+1}} \underbrace{\sum_{\sigma \in T} g_{\sigma}(t) (1-t)^{d - \dim \sigma}}_{\text{polynomial deg} \leq d+1}$$

Coefficient of t^{d+1} is

$$\sum_{\sigma \in T} (-1)^{d - \dim \sigma} = (-1)^d \sum_{\sigma \in T} (-1)^{\dim \sigma}$$

$$= (-1)^d \tilde{\chi}(T)$$

$$= 0$$

End of proof.

Thm: For any P ,

$$H(A(P^0), t) = \frac{h_d^* t + \dots + h_0^* t^{d+1}}{(1-t)^{d+1}}$$

Sketch of proof: Let T be a integral triangulation of P

$$H(A(P^0), t) = \sum_{\substack{\sigma \in T \\ \sigma \not\subset \partial P}} H(A(\sigma^0), t)$$

$$= \sum_{\substack{\sigma \in T \\ \sigma \notin \partial P}} (-1)^{\dim \sigma + 1} H(A(\sigma), \frac{1}{t})$$

Using Euler characteristic

$$= (-1)^{d+1} H(A(P), \frac{1}{t})$$

Let $\text{ehr}(P, k)$ be the coefficient of t^k in $H(A(P), t)$.

$$\text{ehr}(P, k) = \#(kP \cap \mathbb{Z}^n)$$

Thm: $\text{ehr}(P, k)$ is a polynomial in k .

$$H(A(P), t) = \frac{h_0^* + h_1^* t + \dots + h_d^* t^d}{(1-t)^{d+1}}$$

$$[t^k] \frac{h_i^* t^i}{(1-t)^{d+1}} = [t^{k-i}] \frac{h_i^*}{(1-t)^{d+1}} = \underbrace{h_i^* \binom{k-i+d}{d}}_{\text{polynomial in } k}$$

Let $\text{ehr}(P^\circ, k)$ be the coefficient of t^k in $H(A(P^\circ), t)$

$$\text{ehr}(P^\circ, k) = \#(kP^\circ \cap \mathbb{Z}^n)$$

Thm: (Ehrhart reciprocity)

$$\text{ehr}(P^\circ, k) = (-1)^d \text{ehr}(P, -k)$$

Proof sketch: Write out both sides in terms of binomial coefficients, use

$$\binom{-k}{n} = \binom{k+n-1}{n}$$

$$H(\lambda(p), t) = \frac{h_0^* + h_1^* t + \dots + h_d^* t^d}{(1-t)^{d+1}}$$

Then: $h_i^* \geq 0$